

1.3 Classwork: Arithmetic Sequences (continued) — Markscheme

1. [Maximum mark: 7]

$$u_4 = 17 \text{ and } u_9 = 37.$$

$$\text{(a)} \quad u_1 + 3d = 17 \quad \text{and} \quad u_1 + 8d = 37 \quad \text{M1}$$

$$\text{Subtracting: } 5d = 20, \text{ so } d = 4 \quad \text{A1}$$

$$u_1 = 17 - 3(4) = 5 \quad \text{A1}$$

M1 is for any valid route to the two unknowns: writing both equations, using $d = \frac{u_9 - u_4}{9 - 4}$, or solving the system on the GDC. Any of these earns the method mark.

A1 for $d = 4$ and A1 for $u_1 = 5$, awarded independently.

$$\text{(b)} \quad u_n = 5 + (n - 1)(4) = 4n + 1 \quad \text{A1}$$

Accept the unsimplified form $5 + 4(n - 1)$.

FT: award A1 for a correct expression formed from the candidate's own u_1 and d from (a).

$$\text{(c)} \quad \text{Set } u_n = 200: \quad 4n + 1 = 200 \quad \text{M1}$$

$$4n = 199, \text{ so } n = 49.75 \quad \text{A1}$$

n is not a positive integer, so 200 is **not** a term of the sequence. R1

Accept the alternative argument: every term $4n + 1$ is odd (the terms are 5, 9, 13, ...), and 200 is even, so 200 cannot be a term. Award M1 for identifying the parity/structure of the terms, A1 for stating that all terms are odd, R1 for the conclusion.

R1 requires the conclusion to be linked to the reason. "No, because n must be a whole number" is sufficient; a bare "no" with no reason earns no R1.

FT: award R1 for a conclusion correctly drawn from the candidate's own value of n (e.g. concluding that 200 is a term if their n came out a positive integer).

2. [Maximum mark: 6]

Row 1 has 18 seats; each row has 3 more than the one in front, so $u_1 = 18$ and $d = 3$.

(a) $u_n = 18 + (n - 1)(3)$ M1

$u_n = 3n + 15$ A1

Accept the unsimplified form $18 + 3(n - 1)$ for full marks.

M1 is for identifying $u_1 = 18$ and $d = 3$ and using the arithmetic n^{th} -term formula, however written.

(b) $u_{12} = 3(12) + 15 = 51$ seats A1

FT: award A1 for the value obtained by substituting $n = 12$ into the candidate's own expression from (a).

(c) $3n + 15 > 65$ M1

$3n > 50$, so $n > 16.666\dots$ A1

The first such row is row 17. A1

Accept an equation in place of the inequality ($3n + 15 = 65$, $n = 16.67$) provided the candidate then selects the next whole row.

Accept a table search: full marks for showing both adjacent rows, $u_{16} = 63$ (≤ 65) and $u_{17} = 66$ (> 65), and stating row 17. A single row ($u_{17} = 66$, so row 17) still earns full marks, with the feedback note that the adjacent row u_{16} is what justifies “first”. See markscheme-spec §18.

16.666... is not an acceptable final answer — the answer must be a row number.

FT: award the final A1 for the smallest integer n correctly read off from the candidate's own boundary value.